

Coding & Solution

Date	1 July 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/DURGAGANGADHAR18/Emotion-Detection-Learning-Support-Engine.git
Programming Language(s)	Python
Framework(s) Used	Streamlit, TensorFlow, Keras
Key Features Implemented	BERT and Multi-label BiLSTM emotion detection, primary and mixed emotion detection with confidence scores, model comparison, Google Gemini AI powered learning guidance, personalized study plan generation, session and CSV history storage, emotion and subject analytics dashboard, downloadable AI report
Pending / Incomplete Features	User authentication, multilingual emotion detection, voice/facial emotion recognition, LMS integration, cloud deployment
Setup / Run Instructions	1. Clone repository. 2. Create virtual environment. 3. Install dependencies using pip install -r requirements.txt. 4. Add your Google Gemini API key to a .env file. 5. Run the application using streamlit run app.py.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture with separate modules for preprocessing, BERT prediction, BiLSTM prediction, Gemini AI guidance generation, and history/analytics management. The application was tested across emotion prediction, model comparison, guidance generation, and CSV storage workflows, and all core features function as expected. The project is fully functional and ready for further enhancements such as multilingual support and cloud deployment.